PatentForge

AI-Powered Patent Landscape Research Tool

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PatentForge is a research tool, not a legal service. The author of this tool is not a lawyer. The AI systems that execute these prompts are not lawyers. No attorney-client relationship is created by using this tool. See LEGAL_NOTICE.md for full details.

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https://github.com/scottconverse/patentforge

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1. Overview

PatentForge is an open-source, self-hosted web application that helps inventors explore the patent landscape for their ideas using AI. It runs a 6-stage analysis pipeline using Anthropic's Claude API to restate your invention in technical language, search for related patents, map it against patent law requirements, and assemble everything into a structured report you can take to a patent attorney.

The goal is preparation, not replacement. PatentForge helps you walk into your first attorney meeting with your invention clearly described, related prior art identified, and the right questions already on the table. It does the homework so you can make the most of your attorney's time.

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2. What PatentForge Does

- **Structured technical analysis** - 6-stage AI pipeline restates your invention in patent terminology, searches for related art, identifies potential issues, and organizes findings into a structured report.
- **Prior art discovery** - automated patent search via PatentsView API with AI-powered query extraction and relevance scoring, plus Claude web search for papers, products, and open-source projects.
- **Cost transparency** - pre-run cost estimate with per-stage token tracking so you know what the AI processing will cost before you start.
- **Resume from interruption** - pick up where you left off if a run stops mid-pipeline.
- **Multiple export formats** - HTML, Word (.docx), and Markdown for sharing with your attorney.
- **Self-hosted & private** - runs on your machine; invention data stays local except for Anthropic API calls.
- **Legal guardrails** - clickwrap agreement, embedded disclaimers, watermarked exports, dual license for prompt content.

3. System Architecture

PatentForge uses a three-service federated architecture. Each service is independently deployable and communicates via HTTP requests and Server-Sent Events (SSE) for real-time streaming. SQLite is used for local development; PostgreSQL for Docker deployment.

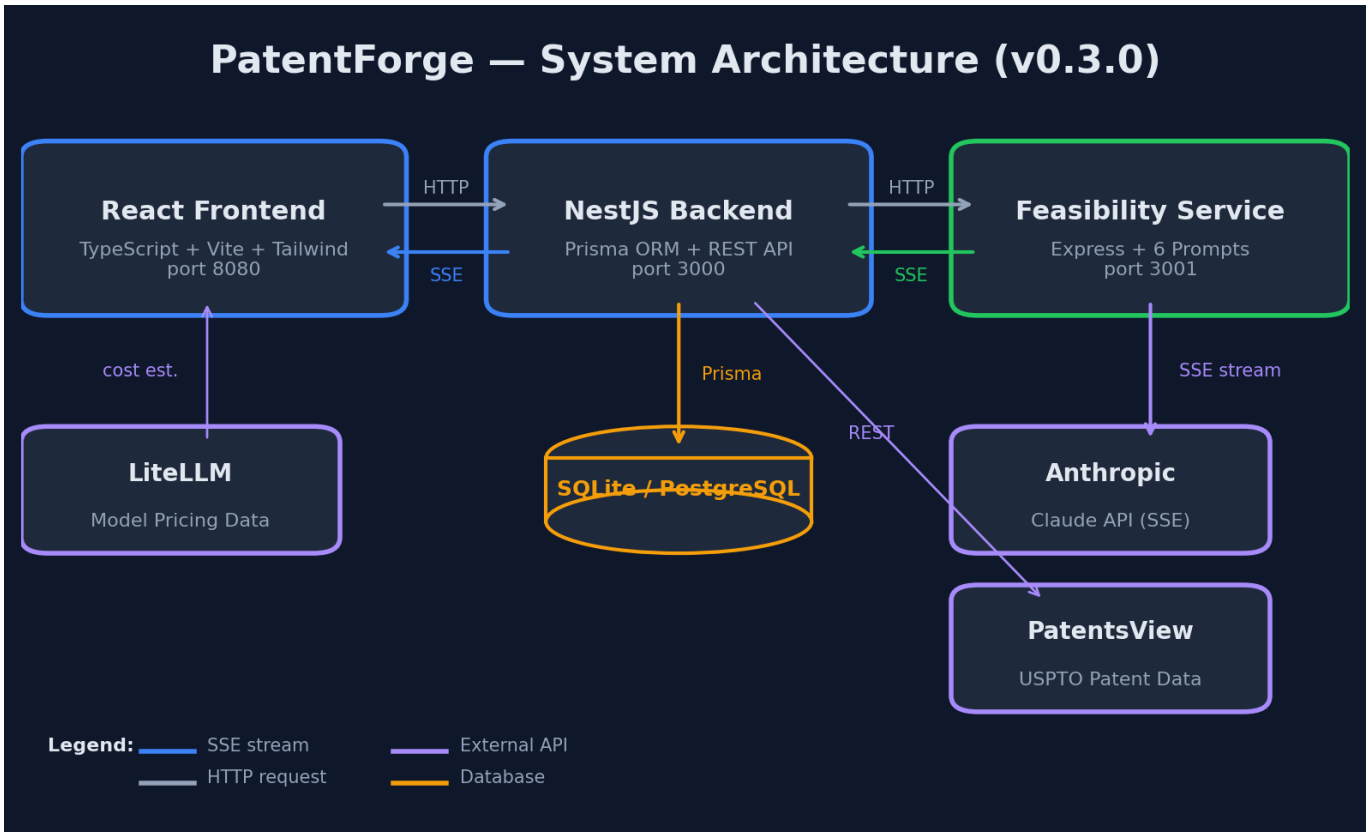

Figure 1: PatentForge System Architecture (v0.3.0)

Service Descriptions

- **React Frontend (port 8080)** - Invention intake form, real-time streaming output with stage progress indicators, report viewer with export buttons. Built with React 18, TypeScript, Vite, and Tailwind CSS.
- **NestJS Backend (port 3000)** - Project CRUD, settings management, feasibility run tracking, SSE event forwarding, prior art search via PatentsView API, Word (.docx) export generation. Built with NestJS, Prisma ORM, SQLite/PostgreSQL.
- **Feasibility Service (port 3001)** - 6-stage sequential AI analysis pipeline. Each stage uses a markdown prompt template with the Anthropic Claude API. Streams tokens in real time via SSE. Built with Express and TypeScript.

External Services

- **Anthropic Claude API** - LLM processing with web search tool for grounded research in Stages 2, 3, and 4.
- **PatentsView API** - USPTO patent database search for automated prior art discovery.
- **LiteLLM Pricing** - Dynamic model cost estimation with 1-hour cached pricing data.

4. The 6-Stage Research Pipeline

PatentForge runs a sequential 6-stage analysis pipeline. Each stage builds on the output of all previous stages. The pipeline streams tokens in real time so you can watch the AI write its findings as they are generated.

Stage 1: Technical Intake & Restatement

Restates your invention in precise technical language. Identifies core components, inventive concept candidates, and CPC classifications. Flags information gaps.

Stage 2: Prior Art Research

Searches for existing patents, academic papers, products, and open-source projects using Claude web search and PatentsView data. Produces an element-by-element comparison table and white space assessment.

Stage 3: Patentability Assessment

Maps the invention against patent law requirements under 35 USC 101 (eligibility), 102 (novelty), 103 (obviousness), and 112 (enablement). Identifies anticipated examiner concerns.

Stage 4: Deep Dive Analysis

Domain-specific analysis of the technology areas relevant to the invention. Patent landscape, strongest/weakest elements, claim framing considerations.

Stage 5: IP Landscape Assessment

Presents filing indicators, recommended IP protection mix, filing strategy, cost estimates, freedom-to-operate flags, and an overall assessment.

Stage 6: Consolidated Report

Assembles all findings into a single structured document with executive summary, risk scores, and a plain-English summary for non-technical readers.

5. Data Flow

The diagram below shows how data flows through the PatentForge pipeline, from inventor input through the 6 analysis stages to the final exported report.

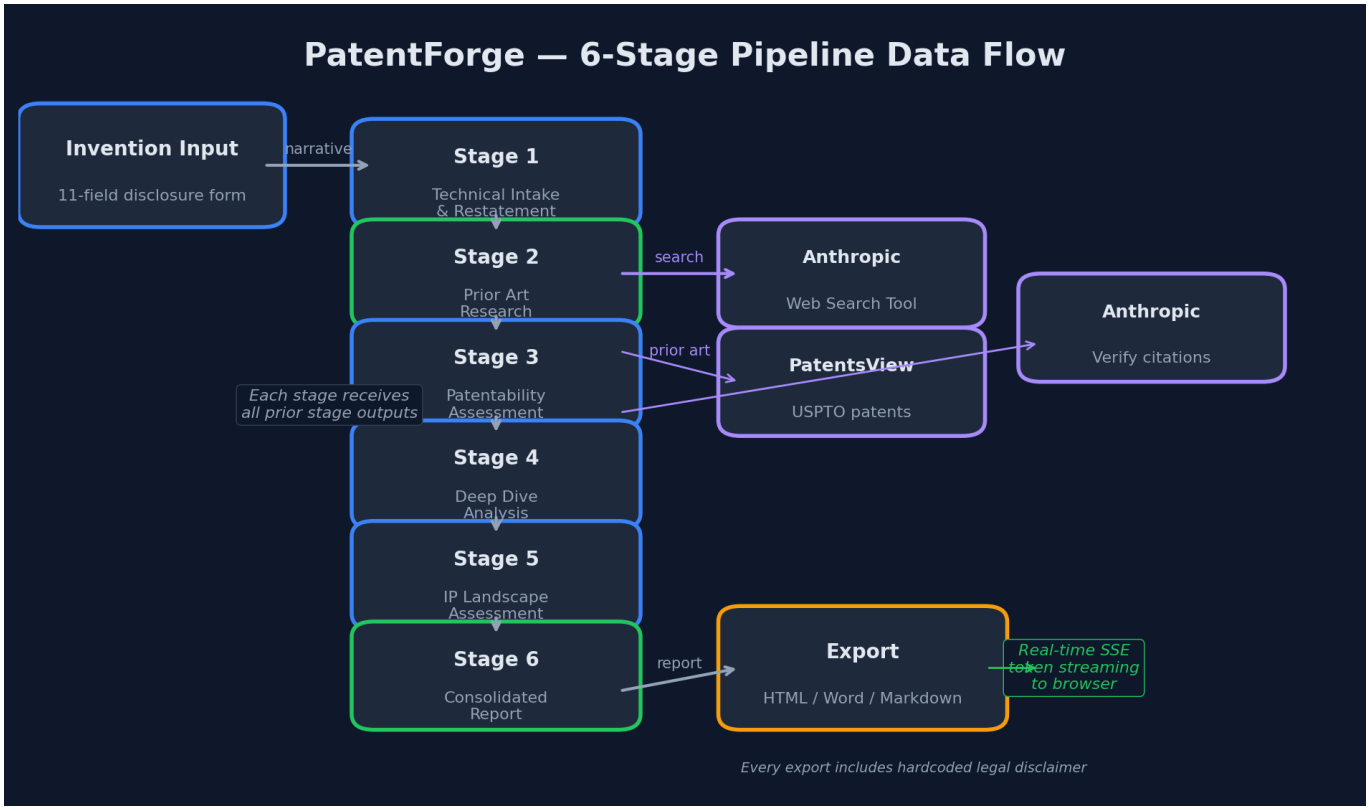

Figure 2: Cross-Stage Data Flow

Key characteristics: each stage receives the full output of all prior stages as context. Stages 2, 3, and 4 use Anthropic web search for grounded research. The backend also queries PatentsView independently for verified USPTO patent data that feeds into Stage 2. Every export (HTML, Word, Markdown) includes a hardcoded legal disclaimer footer that persists even if the AI output is truncated.

6. Database Schema

PatentForge uses Prisma ORM with SQLite for local development and PostgreSQL for Docker deployment. The schema tracks projects, invention disclosures, feasibility runs with per-stage outputs, prior art searches, and application settings.

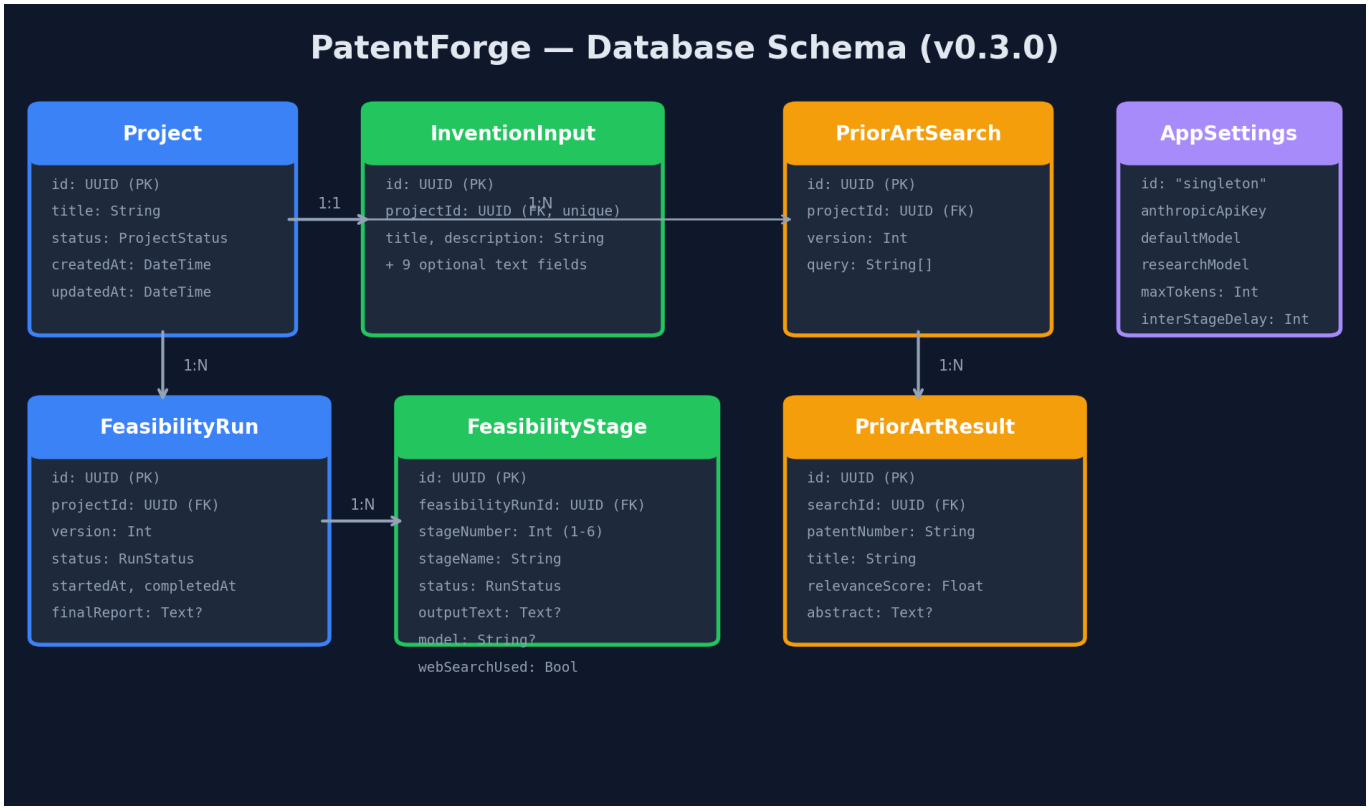

Figure 3: Database Entity-Relationship Diagram

7. Docker Deployment

PatentForge provides a Docker Compose configuration for single-command deployment. The compose file starts four containers: frontend, backend, feasibility service, and PostgreSQL. A persistent volume stores database data across restarts.

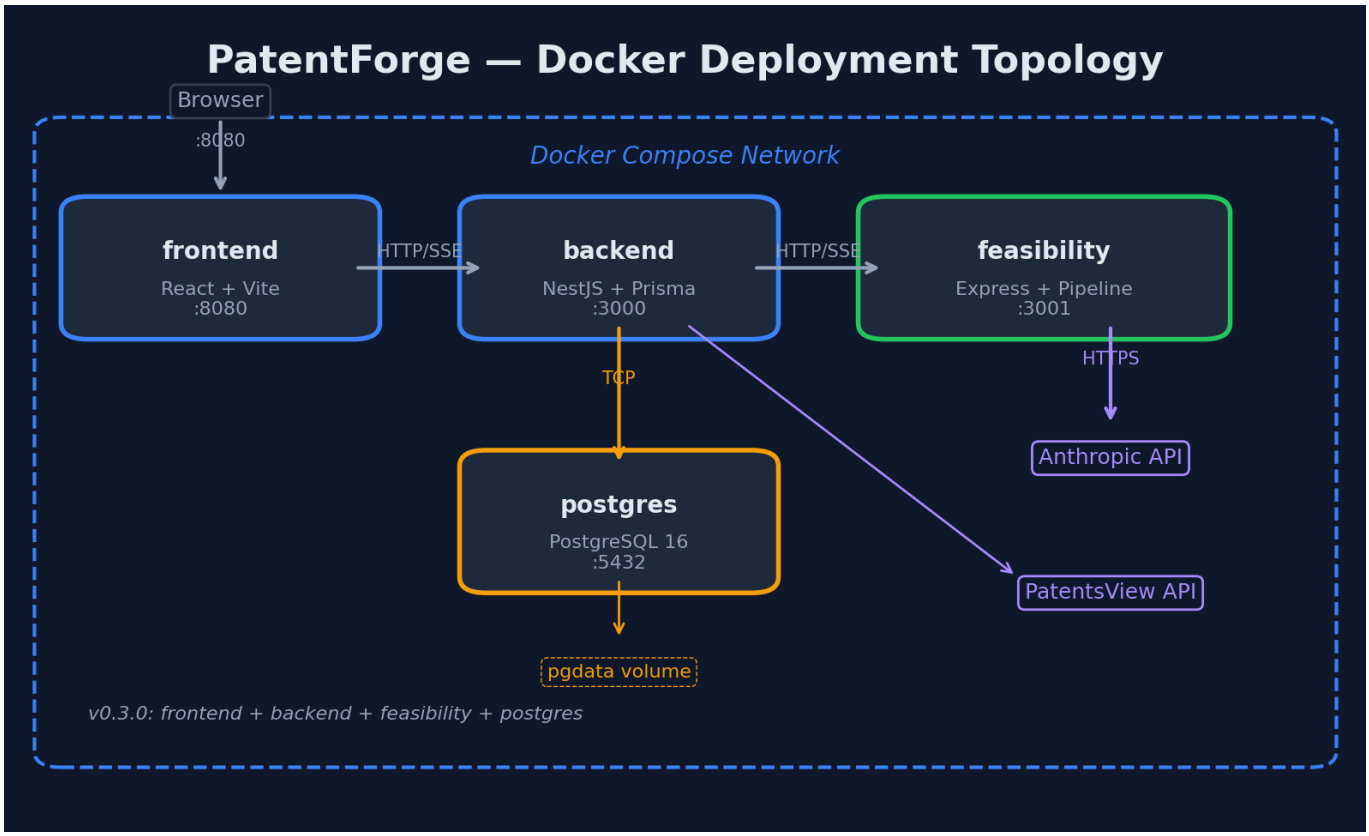

Figure 4: Docker Deployment Topology

8. User Journey

The typical user journey takes an inventor from initial idea through structured research to a prepared attorney consultation.

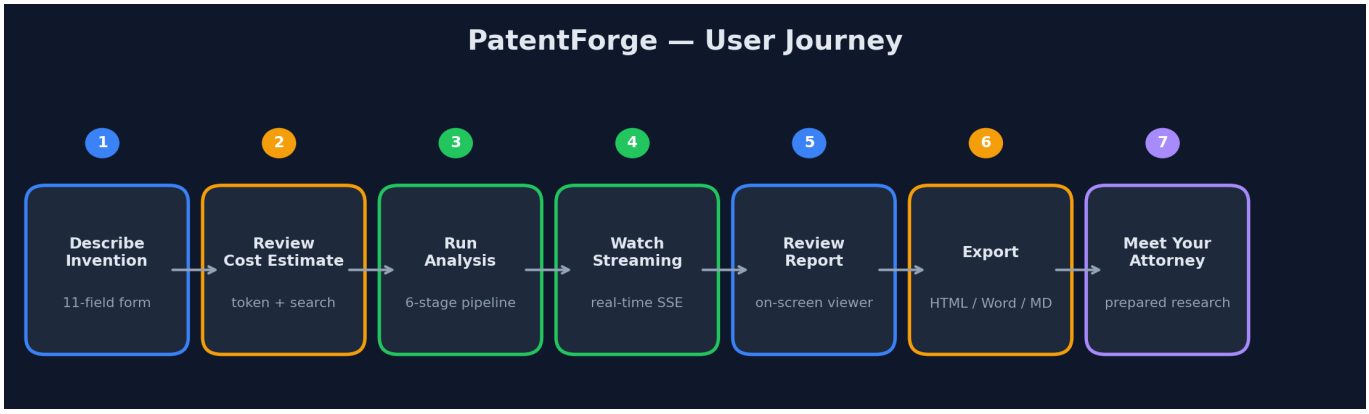

Figure 5: User Journey from Invention to Attorney Meeting

9. Configuration

All settings are configurable via the Settings page in the web UI:

- **Anthropic API Key** - Required. Your Claude API key (BYOK model).
- **Default Model** - Model for most pipeline stages. Default: claude-sonnet-4-20250514.
- **Research Model** - Optional cheaper model for Stage 2 (e.g., Haiku).
- **Max Tokens** - Maximum tokens per stage response. Default: 32,000.
- **Inter-Stage Delay** - Pause between stages for rate limit protection. Default: 5 seconds.

10. Legal Guardrails

PatentForge includes multiple layers of legal guardrails to reduce unauthorized practice of law (UPL) exposure:

- **First-run clickwrap** - unskippable agreement on first launch acknowledging the tool provides research, not legal advice.
- **AI role framing** - all 6 stage prompts identify the AI as a "research assistant" (not an attorney) with explicit "not a lawyer" and "not legal advice" disclaimers.
- **Embedded per-stage disclaimer** - every stage output begins with an italic disclaimer notice that survives copy-paste.
- **Export watermarks** - all generated reports (HTML, Word, on-screen) include a persistent legal disclaimer.
- **Hardcoded HTML footer** - the report exporter includes a disclaimer div outside the AI content that persists even if AI output is truncated.
- **API key disclaimer** - Settings page notes the user is connecting to their own Anthropic account.
- **Evidence-based framing** - output uses "indicators" and "assessment" language rather than prescriptive legal advice.

11. License

PatentForge uses a dual license structure:

- **Code** (backend, frontend, services infrastructure): MIT License
- **Prompt content** (services/feasibility/src/prompts/): Creative Commons Attribution-ShareAlike 4.0 International (CC BY-SA 4.0)

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